

BACHELOR OF SCIENCE EXAMINATION, 2024

(First Year, First Semester)

MATHEMATICS

PAPER : Stat – I (Minor)

(Probability and Descriptive Statistics)

Full Marks : 32

Time : 1 Hour 30 Minutes

Use a separate Answer- Script for each part

PART—I

(Marks : 10)

Answer *any two* questions :

$5 \times 2 = 10$

1. (a) Derive the linear regression equation of x on y . Show that it passes through a fixed point.
- (b) Out of two regression lines $x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$, which one is the regression line of x on y ? (3+2)
2. (a) Find the Spearman's rank correlation coefficient when there are no ties.
- (b) Show that Spearman's rank correlation coefficient (with no ties) for perfect disagreement. (3+2)

(2)

3. If $u = x \cos \theta + y \sin \theta$, $v = y \cos \theta - x \sin \theta$, and u and v are uncorrelated for n pair of values (x_i, y_i) , $\forall i = 1(1)n$, then

prove that $\tan 2\theta = \frac{2r_{xy}s_x s_y}{s_x^2 - s_y^2}$, $s_x^2 =$ variance of x . (5)

PART—II

(Marks : 22)

Answer question number 1 and *any four* from the rest.

1. Suppose X is a real valued random variable with $E(X) = 5$ and $\text{var}(X) = 4$. Find the values of (a) $\text{var}(3X - 2)$
(b) $E(X^2)$ (2)

2. Suppose that the two-dimensional random variable (X, Y) has the probability density function f given by

$$f(x, y) = \begin{cases} 6x^2 y & \text{For } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal probability density function of X (2)

- (b) Find the marginal probability density function of Y (2)

- (c) Are X and Y independent? (1)

(3)

3. If A and B are stochastically independent events such that $P(A) = 0.37$, and $P(B) = 0.44$, then find the values of
(a) $P(A')$ (b) $P(A \cup B)$ (c) $P(A \cap B)$ (d) $P(A \cap B')$
(e) $P(A' \cap B')$. (5)

4. A random variable X can assume the values 1, 2, 3, 4 with probabilities $1/4, 1/2, 1/8, 1/8$ respectively. Determine the distribution function and draw the graph. (5)

5. If the distribution function of a random variable X is given by

$$F(x) = \begin{cases} 1 - (1+x)e^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

find

- (a) $P(X \leq 2)$, $P(1 < X < 3)$ and $P(X > 4)$ (3)
(b) the density function of X (2)

6. (a) If X is a real valued random variable such that $\mu = E(X)$ and $\sigma^2 = \text{var}(X)$, then prove that

(i) $\text{skew}(X) = \frac{E(X^3) - 3\mu\sigma^2 - \mu^3}{\sigma^3}$ (1)

(ii) $\text{kurt}(X) = \frac{E(X^4) - 4\mu E(X^3) + 6\mu^2\sigma^2 + 3\mu^4}{\sigma^4}$ (2)

(4)

- (c) A standard, fair die is thrown and the score X is recorded.
Find skew (X). (2)

